Ogbonna Arinze

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SUMMARY

Senior Frontend Engineer with 8 years of experience building high-performance, conversion-focused web applications across fintech, healthcare, and enterprise environments. Deep expertise in **React**, **TypeScript**, and **Next.js**, with proven success delivering high-traffic B2C frontends that process millions of daily transactions. Experienced in frontend-backend API integration with **NodeJS** and **AWS** services (CloudFront, S3), accessibility-first design systems, performance optimization through code splitting and CDN strategies, and end-to-end product ownership under a "you build it, you run it" philosophy.

CORE COMPETENCIES

- **Frontend Engineering:** React, Next.js (SSR/SSG/ISR), TypeScript, JavaScript (ESNext), Vue.js, Redux, Zustand, React Query, and responsive UI development.
- **Performance & Optimization:** Core Web Vitals, lazy loading, code splitting, CDN asset delivery, Lighthouse, bundle analysis (Webpack, Rollup), and caching strategies.
- **Design Systems & Accessibility:** Radix UI, Tailwind CSS, Ant Design, Storybook, WCAG compliance, and reusable component architecture.
- **Backend & API Integration:** NodeJS REST APIs, NestJS, JWT authentication, RBAC, real-time streaming (SSE, EventSource), and frontend-backend data flow.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, GitHub Actions, CI/CD pipelines, and automated deployment workflows.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, ESLint, Prettier, Husky, and CI-driven end-to-end testing.
- **Data Visualization:** Recharts, react-chartjs, dynamic dashboards, and real-time data streaming interfaces.
- **Leadership:** Technical direction, architecture reviews, mentorship, Agile/Scrum, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension, optimizing workflows for clinical site integrations and enhancing user adoption.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, **accessibility (WCAG)**, and developer velocity across multiple product teams.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Co-developed a full-stack **Next.js** chat widget powered by a RAG system using OpenAI GPT models, **LangChain**, Vercel AI SDK, and Chroma vector database, enabling users to query their data in real time via a conversational interface.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%**, directly impacting revenue for financial institution clients.
- Developed critical payment processing modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing conventional socket.io methods for faster data updates and a more responsive, high-traffic interface.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design** UI.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices with Product Owners.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

Founders Lair Beijing, China (Remote)

Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, collaborating with the business team to align technical solutions with product needs.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to Heroku and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria

Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng

Senior Engineer *Feb 2026 - Present*

AI-powered civic tech platform for Nigerian budget analysis. 700+ documents, 708K+ vector embeddings, 37 states.

- Architected the platform as an **Nx monorepo** managing 6 applications including a **Next.js** citizen-facing web app with **React 19**, **Tailwind CSS v4**, 22 interactive chart types (Recharts), responsive mobile layouts, and **SSR** for SEO optimization.
- Implemented **SSE streaming** for real-time responses with keepalive heartbeats, phone OTP authentication with session cookies, and URL state management with nuqs.
- Configured **CI/CD** with **GitHub Actions**, **Playwright** E2E tests across 4 projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, and memory limits.

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*